

A note on adversary running times

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Abstract

In this note, we consider a perspective on adversary running times that fixes the adversary's running time to be 2^κ , and then asks: to achieve security against such an adversary, what running time in terms of κ do honest parties need? This perspective gives rise to a new natural class of adversary running times that we call *quasi-exponential* time adversaries.

In cryptography, we often consider the running time of adversaries, and we talk about what class of adversarial running times a particular cryptographic scheme can handle. In this note, we consider a slightly different perspective, that gives rise to at least one more natural class of adversary running times that we name *quasi-exponential*.

What we suggest is that we fix the adversary's running time to be 2^κ , and then ask: in terms of this parameter κ , what is the running time of the honest parties? Informally, to achieve “ κ bits of security” with regard to adversary running time, how long do honest parties have to run for?

Throughout, we use the standard asymptotic shorthand $\text{poly}(n) = n^{O(1)}$. We also use $\text{quasipoly}(n) = n^{\log^{O(1)} n}$ (equivalently, $2^{\log^{O(1)} n}$). When we say that an algorithm runs in sub-exponential time in n , we mean 2^{n^δ} for some constant $\delta > 0$.

Having fixed the adversary's running time to 2^κ , we now consider three cases.

1. Honest parties run in time $\text{poly}(\kappa)$. First, we observe that if the running time of the honest parties is $\text{poly}(\kappa)$, while the adversary is fixed to 2^κ running time, then this corresponds to the class of *sub-exponential adversaries* (in the usual security parameter). Indeed, if the natural security parameter is n and the running time of the honest parties is some polynomial in n , then we may set $\kappa = n^\varepsilon$ for a suitable constant $\varepsilon > 0$ while maintaining polynomial time for the honest parties. Under this translation, the adversary's bound 2^κ becomes

$$2^\kappa = 2^{n^\varepsilon},$$

which is sub-exponential in n .

2. Honest parties run in time $\text{quasipoly}(\kappa)$. Next, we observe that if we relax the running time of the honest parties from $\text{poly}(\kappa)$ to $\text{quasipoly}(\kappa)$, while the adversary is fixed to 2^κ running time, then this gives rise to a new but natural class of adversary running times, which we call *quasi-exponential*. Concretely, in terms of a natural security parameter n (where honest parties run in $\text{poly}(n)$ time), we may take

$$\kappa = 2^{\log^\varepsilon n} \quad \text{for a suitable constant } \varepsilon > 0,$$

so that $\text{quasipoly}(\kappa)$ is now $\text{poly}(n)$. Under this translation, the adversary's time bound becomes

$$2^\kappa = 2^{2^{\log^\varepsilon n}},$$

which we term *quasi-exponential* in n .

3. Honest parties run in sub-exponential time in κ . Finally, we observe that if we relax the running time of the honest parties all the way to be sub-exponential in κ , while the adversary is fixed to 2^κ running time, then this corresponds to *quasi-polynomial security* in the natural parameter n . Namely, taking $\kappa = (\log n)^{1+\varepsilon}$ for some constant $\varepsilon > 0$ can transform any sub-exponential-in- κ honest running time to $\text{poly}(n)$, while the adversary's bound becomes

$$2^\kappa = 2^{(\log n)^{1+\varepsilon}} = (2^{\log n})^{\log^\varepsilon n} = n^{\log^\varepsilon n},$$

which is quasi-polynomial in n .

These three cases suggest that the lens of “fixing the adversary's running time to 2^κ and measuring honest party running time in terms of κ ” organizes several familiar regimes, and also gives rise to the intermediate natural notion of quasi-exponential adversary running times, which can be seen as “nearly sub-exponential” in the same sense that quasi-polynomial algorithmic running times can be seen as “nearly polynomial.”

Composition behavior of quasi-exponential time. Indeed, we can try to investigate the notion of quasi-exponential time by means of its behavior under composition. We know that both polynomial time and quasi-polynomial time are closed under composition. Informally,

$$\text{poly}(\text{poly}(n)) = \text{poly}(n) \quad \text{and} \quad \text{quasipoly}(\text{quasipoly}(n)) = \text{quasipoly}(n).$$

In contrast, when we consider sub-exponential time, if we let:

$$f(n) = 2^{n^\delta} \text{ and } g(n) = 2^{n^\varepsilon}$$

for constants $\delta, \varepsilon > 0$, then we have

$$f(g(n)) = 2^{(2^{n^\varepsilon})^\delta} = 2^{2^{\delta n^\varepsilon}} = 2^{2^{\Theta(n^\varepsilon)}},$$

which is exponential in a sub-exponential quantity (i.e., it lies in an $\text{exp}(\text{subexp})$ regime).

Similarly, when we consider quasi-exponential time, if we let:

$$f(n) = 2^{2^{(\log n)^\delta}} \text{ and } g(n) = 2^{2^{(\log n)^\varepsilon}}$$

for constants $\delta, \varepsilon > 0$, then we have

$$f(g(n)) = 2^{2^{\left(\log\left(2^{2^{(\log n)^\varepsilon}}\right)\right)^\delta}} = 2^{2^{\left(2^{(\log n)^\varepsilon}\right)^\delta}} = 2^{2^{2^{\delta(\log n)^\varepsilon}}} = 2^{2^{2^{\Theta((\log n)^\varepsilon)}}},$$

which again adds one more outer exponentiation, yielding a bound that is exponential in a quasi-exponential quantity (i.e., it lies in an $\text{exp}(\text{quasiexp})$ regime). Thus, we see that quasi-exponential time behaves quite similarly to subexponential time.

Discussion and Context. This note can be seen as putting forward a “theory analog” of the kind of discussions that are commonplace in practice-oriented applied cryptography work, where honest party running times needed to achieve desired levels of security are often reported. We replace such concrete honest party running times with honest party complexity classes, and then observe the adversary complexity classes that arise when honest party running times are scaled back down to polynomial-time.

Acknowledgements. We thank Yuval Ishai and Isaac Hair for their helpful feedback!